

Determination of Quinine in a Soda Sample via Fluorescence Spectrophotometry

Abstract:

A soda sample was analyzed to determine the quinine concentration using fluorescence spectrophotometry. A primary standard was prepared and measured to determine an excitation wavelength. Diluted standards were prepared and measured in the optimized wavelength ($\lambda_{\text{emission}} \approx 450\text{nm}$). A calibration curve relating fluorescence intensity to quinine concentration was generated, yielding a linear relationship $I = kP_0C$. Using the line of best fit, the unknown concentration was calculated to be 10.51mg/L quinine, deviating significantly from the true value of 68.4 mg/L.

Introduction:

Quinine is a bitterant found in tonic water and similar drinks such as colas, responsible for its bitter taste. It is also a 400-year-old drug that is still an ingredient in medications used to treat diseases such as malaria (1). Quinine is a naturally fluorescent compound that absorbs energy when exposed to UV light, then emits this energy as light of a longer wavelength.

$\text{Quinine } (Q) + hv \rightarrow \text{Quinine}^* \rightarrow \text{Quinine } (Q) + hv_2$

In this model, quinine absorbs energy from UV light (hv) and becomes excited to a higher energy state (Quinine^*). As it returns to the ground state, energy is emitted as light (hv_2) at a longer wavelength (2). The light intensity emitted, which is directly proportional to the sample's concentration, can be measured using fluorescence spectrophotometry. Under low concentrations, this proportionality holds, where fluorescence intensity depends linearly on the concentration.

The equation $I = kP_0C$ represents the linear relationship between analyte concentration and fluorescence intensity. I is the measured fluorescence intensity, k is a constant that includes

the sensitivity of the instrument and the efficiency where quinine fluoresces. P_0 is the power of the UV light, and C is the concentration of quinine. Both k and P_0 are constants, meaning that I is directly proportional to the concentration of C . From this relationship, the creation of a calibration curve in the form $y = mx + b$ establishes a standard plot using known concentrations; the concentration of an unknown substance containing quinine can be determined.

Fluorescence spectrophotometry detects very low concentrations of an analyte. It is also highly selective, so only compounds that fluoresce are detected, reducing interference from other compounds. The sample remains intact after analysis, and it is fast and reproducible. The fluorescence spectrophotometer also has drawbacks, as it is limited by the analyte. Not all analytes fluoresce naturally, limiting the main function of the machine. Impurities from outside sources, including air and exposure to light, can affect the results of certain analytes.

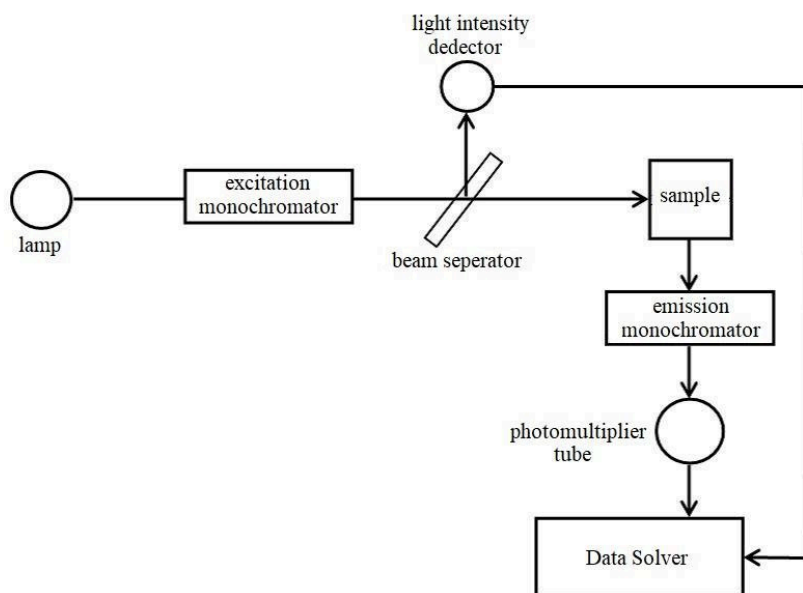


Figure 1: Block Diagram of a Fluorescence Spectrophotometer (3)

The fluorescence spectrophotometer used for this experiment was the Cary Eclipse Fluorescence Spectrophotometer by Agilent Technologies (4). It works by directing a beam of UV light through a sample contained in a cuvette. Quartz is primarily used for cuvettes, as its optical properties, such as light transmittance, low coefficient of thermal expansion, and chemical resistance, allow for it to be non-interfering with the sample(5). The beam is released from a high-energy light source, in this case a xenon flash lamp (80 Hz), and passes through an excitation monochromator. This isolates the wavelength needed to excite the analyte. The analyte is then excited and emits light at a longer wavelength. The emission monochromator is placed perpendicular (90°) to the excitation beam in order to minimize the detection of any scattered excitation light, reducing error associated with stray light. A beam separator, or emission monochromator, is used to isolate the emitted fluorescent light from any remaining excitation light. The resulting light is detected by a photomultiplier tube, which converts the incoming photons into an electric signal. The signal is sent to the data solver, which translates the signal into an intensity reading. This intensity reading can be utilized to calculate the concentration of the analyte.

Experimental:

A quinine standard was prepared by dissolving approximately 12mg of quinine sulfate in 0.1 N sulfuric acid and diluting to 1 L. Dilutions were then performed to create a series of standard solutions with decreasing concentrations. A soda sample was prepared by diluting 1 mL of soft drink in a 100 mL volumetric flask. The optimal excitation wavelength for quinine was then determined by UV spectrophotometry of the standard solution. Fluorescence spectrophotometry was then employed to measure the fluorescence intensity (absorption) of the standards and the soda sample. The absorption of the standards and concentration were then used

to generate a calibration curve, which allowed for the determination of the quinine concentration in the soda sample.

Results:

A linear calibration curve was generated from the absorption values vs. the known concentration of the quinine standards. After applying the equation generated from the calibration plot to the absorbance of the unknown sample, a quinine concentration of 10.51mg/L was determined.

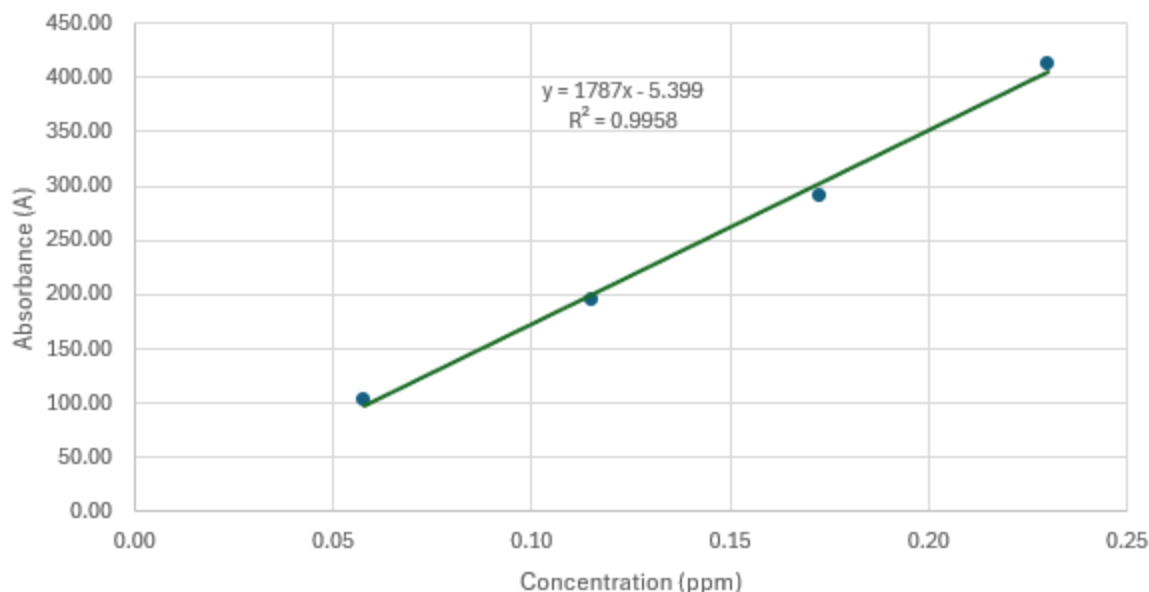
Standard	Dilution Factor
Q25	1:3
Q50	1:1
Q75	3:1
Q100	—

Table 1: Dilutions of Quinine Standards

Analyte	Concentration (ppm)	Emission Intensity (450 nm)
Q100	0.2300	413.17
Q75	0.1725	292.35
Q50	0.1150	196.74
Q25	0.0575	103.10
Sample	0.1051	182.46

Table 2: Concentrations of Quinine Standards and Fluorescence Measurement Values

(450nm Emission)



Graph 1: Absorbance (A) vs. Concentration (ppm) of Quinine in Soda Sample

To determine the concentration of quinine in the soda sample, several calculations were performed. The stock solution was prepared by dissolving 11.5mg of quinine sulfate in 0.5 L of 0.1 N sulfuric acid, resulting $11.5\text{mg} * \frac{1}{0.5}\text{L} = 23\frac{\text{mg}}{\text{L}}$ (23 ppm) of quinine sulfate. Then, 5 mL of this stock solution was diluted to a total volume of 500 mL, yielding the Q100 standard, with a calculated concentration of $\frac{23\text{ppm}(5.00\text{mLQ})}{500\text{mL}} = 0.23\text{ppm}$. Finally, the Q100 was further diluted to prepare three other standards: Q75, Q50, and Q25. These standards used dilution ratios of 3:1, 1:1, and 1:3 (Q100 to 0.1N sulfuric acid). After graphing the intensity from the fluorospectrophotometer, the linear equation $y = 1787x - 5.399$ was established. Using the intensity value of 182.46, the concentration (x) was calculated to be $1.05 * 10^{-1}$. The concentration can be calculated through $M_1V_1 = M_2V_2$, M1 is the x value, V1 is the dilution factor of the analyte or 100mL, M2 is the concentration of the unknown, and V2 is the volume of soda used $0.105 * 100\text{mL} = M_2(1\text{mL})$. After solving, the concentration of M2 is 10.5125 mg/L quinine in the soda sample.

Discussion and Conclusion:

The experimental results indicate a measured quinine concentration of approximately 10.51mg/L in the soda sample, compared to the true value of 68.4 mg/L of quinine. The deviation from these values is approximately 84.7%, indicating systematic error that significantly impacted the results from this experiment. The relative error is approximately 579 parts per thousand (ppt) lower than the true value, further confirming the influence of a systematic error.

Several factors may have contributed to the systematic error present in the experiment. The most probable source of error is the dilution process that occurred when preparing the standards. When handling small volumes and performing several dilutions, minor inaccuracies amplify after each dilution. Another factor, quenching due to air exposure, resulting from the delay between dilution and measurement, likely affected the results. Calibration or instrumentation errors could have also influenced the results if present.

The fluorospectrometric method used in this experiment offers several advantages when paired with the correct compound. If the analyte is the only fluorescent compound present at a given wavelength, the analyte can be measured with minimal background interference. Fluorescence spectrophotometers do not require a large amount of sample to measure their concentration, only enough to fill a cuvette. Some drawbacks are that fluorospectrophotometers are extremely sensitive to changes in factors such as pH. Many fluorescent molecules are also prone to quenching from air, meaning they need to be sealed or tested swiftly after synthesis. It is only applicable to molecules that can fluoresce, as this is what the machine is measuring.

Alternative quantitative methods for measuring quinine in a sample are also widely accepted for their validity. UV spectrophotometry can be used, but this requires further isolation of the analyte and may introduce more background interference. Mass spectrometry can also be

used to measure quinine, but this methodology, while improving in accuracy, is much more expensive to operate (6).

Overall, the indication of a systematic error signals the need for further trials to validate the quinine concentration of the sample. As there was a reading, quinine was likely present in all samples tested, but may have been reduced due to an incorrect dilution or outside error.

Bibliography

- (1) Achan, J.; Talisuna, A. O.; Erhart, A.; Yeka, A.; Tibenderana, J. K.; Baliraine, F. N.; Rosenthal, P. J.; D'Alessandro, U. Quinine, an Old Anti-Malarial Drug in a Modern World: Role in the Treatment of Malaria. *Malaria Journal* 2011, 10.
<https://doi.org/10.1186/1475-2875-10-144>. (accessed 2025-04-06)
- (2) Lower, S. 9.8: *Molecular Orbital Theory*. Chemistry LibreTexts.
[https://chem.libretexts.org/Bookshelves/General_Chemistry/Chem1_\(Lower\)/09%3A_Chemical_Bonding_and_Molecular_Structure/9.08%3A_Molecular_Orbital_Theory](https://chem.libretexts.org/Bookshelves/General_Chemistry/Chem1_(Lower)/09%3A_Chemical_Bonding_and_Molecular_Structure/9.08%3A_Molecular_Orbital_Theory).
 (accessed 2025-04-05)
- (3) *Sensitive. Accurate. Flexible. Agilent Cary Eclipse Fluorescence Spectrophotometer*.
https://www.agilent.com/cs/library/brochures/5990-7788EN_Cary_Eclipse_Brochure.pdf.
 (accessed 2025-04-05)
- (4) *Quartz Glass: What Is It? How Is It Made? Properties, Uses*. www.iqsdirectory.com.
<https://www.iqsdirectory.com/articles/glass-cutting/quartz-glass.html>. (accessed 2025-04-05)
- (5) Can Akaoğlu. *INVESTIGATION OF FILM FORMING, OPTICAL AND ELECTRICAL PROPERTIES OF AgNPs DOPED PS/AgNPs COMPOSITES*. ResearchGate.
https://www.researchgate.net/publication/322641696_INVESTIGATION_OF_FILM_FORMING_OPTICAL_AND_ELECTRICAL_PROPERTIES_OF_AgNPs_DOPED_PSAgNPs_COMPOSITES (accessed 2025-04-05).
- (6) Merck. NMR Chemical Shifts of Impurities. *Merck* 2025, 1.
<https://www.sigmaaldrich.com/US/en/technical-documents/protocol/analytical-chemistry/>

[thin-layer-chromatography/tlc-determination-of-quinine-in-tonic-water-and-caffeine](#)

(accessed 2025-04-05)